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SUPERSEDING
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17 May 1973

MILITARY SPECIFICATION

GLOVE INSERTS, RADIOACTIVE CONTAMINANTS PROTECTIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This specification covers the requirements for knitted cotton glove inserts, of one type and size only.

2. APPLICABLE DOCUMENTS

*2.1 Issues of Documents The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

V-T-276 - Thread, Cotton

MILITARY

MIL-B-17757 - Boxes Shipping, Fiberboard, (Modular Sizes)
MIL-T-43548 - Thread, Polyester, Cotton-Covered

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-751 - Stitches, Seams, and Stitching

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Navy Cltohing & Textile Research Facility, 21 Strathmore Road, Natick, MA 01760, by using the self-addressed standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8415

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, drawings, standards, and publications required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

US POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendant of Documents, US Government Printing Office, Washington, DC 20402).

2.2 Other publications The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issues in effect on the date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, ATTN: Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, ILL 60606).

3. REQUIREMENTS

3.1 Guide sample Samples, when furnished, are solely for guidance and information the contractor (see 6.3). Variations from this specification may appear in the sample, in which case this specification shall govern.

3.2 First article approval When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3).

3.3 Materials

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*3.3.1 Cloth, cotton, jersey knit The cloth used for the gloves shall be plain (jersey) circular knit, made with singles cotton yarn. The finished fabric shall be scoured and bleached or natural and shall conform to the physical requirements specified in Table I when tested as specified in 4.4.1.

Table I - Physical requirements

Characteristic	Minimum Requirement
Weight oz/sq yd	2.8
Wales / in	24
Courses/in	26

* 3.3.2 Thread The thread for seaming and stitching the gloves shall be cotton, natural or white in color, conforming to type 1A1 or 1A2, ticket no. 70, 2 ply, of V-T-276 or as an alternate thread, polyester, cotton-covered, size 70 conforming to MIL-T-43548 may be used.

3.4 Design The gloves shall be an inseam sewn, 5-finger silhouette, ambidextrous design, as shown on figure 1.

3.4.1 Figure Figure 1 is furnished for information purposes only. To the extent of any inconsistencies between the written specification and the figure, the written specification shall govern.

3.5 Patterns Standard patterns will be furnished by the government to the contractor for use in cutting the working patterns (see 6.3). The working patterns shall be identical to the government patterns. Neither the government patterns nor the working patterns shall be altered in any way.

3.5.1 List of pattern parts The component parts of the gloves shall be cut from the following material as specified and in strict conformance with the number of parts required for the manufacturing process as follows:

Material	Nomenclature of pattern parts	Cut parts/pair
Cloth, cotton jersy knit	Interchangeable face	4

3.6 Construction

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*3.6.1 Stitches, seams, and stitching Stitch, seams, and stitching types specified in Table II shall conform to FED-STD-751. When two or more methods or stitch types are given for the same part of an operation, any one may be used. Unless otherwise specified, all stitching shall finish a minimum of 1/8 inch from the raw edge. Trimmer attachment may be used.

3.6.2 Stitches per inch The minimum and maximum number of stitches per inch shall be as specified in Table II.

*3.6.3 Thread breaks and ends of stitching Ends of all stitching produced with 301 stitch types shall be backtacked not less than 1/4 inch. Both ends of all stitching produced with 401 stitch type shall have a 1/4 to 1/2 inch chain extending beyond each end. Thread breaks (all stitch types) shall be secured by stitching back of break not less than 1/2 inch. Skipped stitches or thread breaks (401 stitch type) may be repaired using 301 stitch type.

3.7 Manufacturing operation requirements The gloves shall be manufactured in accordance with operation requirements specified in Table II. The contractor is not required to follow the exact sequence of operations listed provided that the finished gloves are identical to those produced by following the sequence of operations listed in Table II.

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Table II - Construction of gloves

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
1.	The gloves shall be marked or cut in accordance with the pattern furnished. All component parts shall be cut with the length direction (cuff to fingertip) of the part in the wale direction. NOTE: During the manufacturing operation, components having material defects or damages that are classified as defects in section 4 shall be removed from production and replaced with non-defective components.					
2.	Assemble gloves Join two component parts, 1/4 gauge from outer edges, except at top opening. NOTE: 1/8 inch seam allowance trimmed off during stitching or after stitching operation.	301 or 401	SSa-1	11-13	70/2	70/2
3.	Trimming and cleaning Trim thread ends and surplus material, remove all spots, stains and loose ends.					
4.	Finishing Turn gloves, right side out. Form or lay off gloves on heated forms.					

NCTR Form 1, 15 Sep 1967

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*3.8 Measurements The finished gloves shall conform to the dimensions as specified in Table III. All dimensions and tolerances are expressed in inches. The measurements shall be made at location A, B, and C with the gloves laid flat under no tension.

Table III - Finished measurements

Location of measurement	Dimension	Tolerance
A Overall length	14	± 1/4
B Width across palm	4 1/2	± 1/4
C Width across top opening	3 7/8	± 1/4

NOTE: A, B, and C refer to figure 1.

A measurement taken from tip of second finger to top opening.
B measurement taken across palm at thumb crotch
C measurement taken across the top opening.

3.9 Workmanship The finished gloves shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

*4.2 Classification of inspection The inspection requirements specified herein are classified as follows:

1. First Article inspection (see 4.3).
2. Quality Conformance inspection (see 4.4).

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4.3 First article inspection When required, the first article submitted in accordance with 3.2 shall be inspected for compliance with design, construction, workmanship, appearance, and dimensional requirements as specified in 4.4.2.1 and 4.4.2.2.

4.4 Quality conformance inspection Inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein.

*4.4.1 Component and material inspection In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document. In addition, the finished knitted fabric shall be tested for characteristics specified in section 3 in accordance with referenced test methods specified in Table IV of FED-STD-191. The sample unit shall be 1 yard full width of the finish fabric. The lot size shall be expressed in units of one yard each. The sample size shall be as shown below and the lot shall be unacceptable if one or more units fail to meet any requirement specified. All test reports shall contain the individual value utilized in expressing the final result.

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

Table IV - Fabric tests

Component	Characteristic	Requirement Paragraph	Test Method
Cloth, cotton jersey knit	Material identification	3.3.1	1200 1/
	Type of knit	3.3.1	Visual 2/
	Color	3.3.1	Visual 2/
	Weight	Table I	5041
	Wales per inch	Table I	5070
	Courses per inch	Table I	5070

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.
2/ One determination shall be performed per sample unit and the results shall be reported as "pass" or "fail".

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4.4.2 Examination of the end item The defects found during the examination of the end item shall be classified in accordance with 4.4.2.1 and 4.4.2.2. The applicable inspection level and acceptable quality levels (AQL) shall be as indicated in 4.4.2.3. The sample unit shall be one glove and the lot size shall be expressed in units of one glove.

4.4.2.1 Visual examination The gloves shall be examined for defects classified in accordance with the list below:

Defects	Classification Major	Minor
MATERIAL DEFECTS AND DAMAGES		
a. Any hole, tear, float, smash, drop stitch, mend, or needle chew	X	
b. Any pulled or snagged yarn		X
c. Slubby yarn more than twice the diameter of the normal diameter of yarn being used		X
CLEANNESS		
a. Any permanent spot or stain (excluding pattern printing marks) greater than 1/4 inch in any direction		X
b. Thread end not trimmed		X
c. Loose threads not removed		X
COLOR		
Other than specified		X
CUTTING		
Any part not cut in accordance with specified pattern	X	
SEAMS AND STITCHINGS		
a. Accuracy of stitching		
1. Seam twisted or pleated	X	
2. Ends of stitching produced with a 301 stitch type, backtacked less than 1/4 inch		
3. Ends of stitching produced with 401 stitch type, having chain extended less than 1/4 inch or more than 1/2 inch beyond end		X
4. Ends of stitching produced on 401 stitch type not having a chain extending beyond each end	X	

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Defects	Classification Major	Minor
SEAMS AND STITCHINGS (cont'd)		
5. Thread breaks not repaired properly		X
6. Inseam stitching less than 1/8 inch from edge	X	
b. Any open seam	X	
NOTE: One or more broken stitches or run offs constitute an open seam	X	
c. Seam or stitch type not as specified		X
d. Stitch tension (too tight or too loose)	X	
e. Stitches per inch (scored when the condition exists on more than half the length of seam or stitching)		
1. One stitch more or less than the maximum or minimum specified		X
2. Two stitches more or less than the maximum or minimum specified	X	
ASSEMBLY DETAIL		
a. Not constructed as specified	X	
b. Gloves not layed off or formed on heated forms		X

4.4.2.2 Dimensional examination The gloves shall be examined for dimension in accordance with established tolerances (see 3.8).

4.4.2.3 Inspection levels and acceptable quality levels The inspection levels and acceptable quality levels (AQL's) expressed in defects per 100 units for visual and dimensional examination shall be as follows:

		AQL's	
	Inspection level	Major	Total
For defects applicable to 4.4.2.1	I	2.5	6.5
For defects applicable to 4.4.2.2	S-2	-	4.0

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4.4.3 Examination of packaging requirements An examination shall be made to determine that packaging, packing and marking complies with section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery, except that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per 100 units (DHU).

5. PACKAGING

5.1 Preservation-packaging Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A Twelve pairs of gloves shall be evenly stacked in a bundle with every six pairs reversed end to end. The bundle shall be securely tied approximately three inches from each end with cotton tape or twine.

5.1.2 Level C (Commercial packaging) Gloves shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The packages and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A One hundred and eighty (180) pairs of gloves, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled and closed conforming to class weather resistant, grade V2s, size 1 of MIL-B-17757. Level A packages shall be packed flat, one in length, three in width and four in depth, with the length of the bundles parallel to the side panels of the container. In the remaining space at the opposite end panel, place three bundles on side edge, one in length and three in depth, with the length of the bundles parallel to the end panel. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial kraft paper.

5.2.2 Level B One hundred and eighty (180) pairs of gloves, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled and closed conforming to class domestic, grade 200, size 1 of MIL-B-17757. Level A packages shall be packed flat, one in length, three in width and four in depth with the length of the bundles parallel to the side panels of the container. In the remaining space at the opposite end panel, place three bundles on side edge, one in length and three in depth, with the length of the bundles parallel to the end panel. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial kraft paper.

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5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with MIL-B-17757 and closed in accordance with the appendix of the container specification.

5.2.3 Level C (Commercial packing) Gloves, packaged as specified in 5.1 shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Container shall comply with the US Postal Service Manual, Uniform Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use The gloves covered by this specification are intended for use by personnel working in the presence of low level radioactive contamination as an insert in combination with MIL-G-82242 Glove, Shells, Radioactive Contaminant Protective.

6.2 Ordering data Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. When first article sample is required (see 3.2).
c. Selection of applicable levels of packaging and packing required (See 5.1 and 5.2).
d. When weather resistant grade fiberboard shipping containers are required for level B packaging (see 5.2.2.1).

6.3 Samples and patterns For access to samples and patterns, address the procuring office issuing the invitation for bids.

* 6.4 Changes from previous issue The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, and deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Navy - SA

Preparing Activity:
Navy - SA

Review activities:
DLA - CT
Navy- SH, MS
Air Force - 45

Project No. 8415-0048

User activities:
Navy - CG, AS, OS

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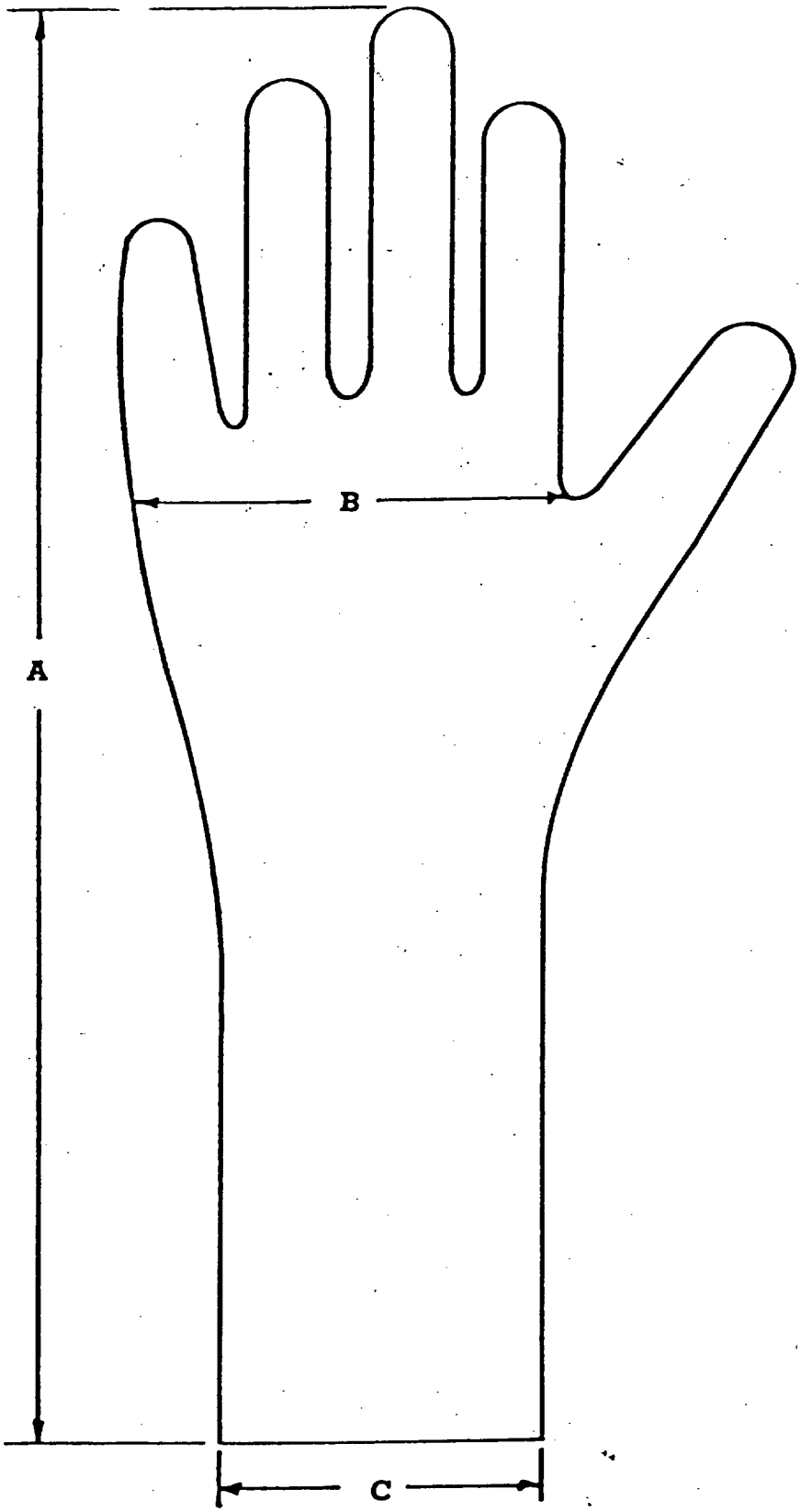


FIGURE 1. Gloves, Inserts, Radioactive Contaminants Protective

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